Agentic Retrieval-Augmented Generation for the Literature on Machine-Learned Interatomic Potentials

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Abstract

[Write last. Four sentences: the problem (answering questions over a specialised literature with verifiable citations), what you built (a citation-graph corpus, a cited-answer pipeline, an agent, and a hand-labelled benchmark), the main result with its uncertainty (e.g. the agent raises correctness from X to Y on N questions, fixing A and breaking B), and the cost of that gain.]

1 Introduction

2 Related work and background

The domain. Machine-learned interatomic potentials approximate the potential-energy surface of atomic systems at close to quantum-mechanical accuracy and a fraction of its cost. The field has moved quickly: from neural networks on fixed, hand-designed descriptors of each atom's environment [2], through message-passing networks that learn their own representations [11], to networks whose internal features are equivariant under rotations [1], and to potentials trained against experimental data rather than quantum reference calculations [13]. Classical biomolecular force fields evolve alongside them [5]. For retrieval this history matters: the same idea is named differently across generations of papers ("symmetry functions" versus "equivariant features"), many answers are equations, and later papers restate earlier results, so a question can be answered by several sources.

Retrieval-augmented generation and agents. Retrieval-augmented generation (RAG) conditions a language model on passages retrieved for the input, which grounds its answers in a corpus and lets them cite their sources [7]. In its basic form retrieval happens once, with the question as the query, so an answer can only be as good as that one search. Agents relax this restriction: ReAct [15] interleaves the model's reasoning with actions such as searches, so that what it reads determines what it looks for next. Our agent follows this pattern, with retrieval and reading as its actions.

Agents for the scientific literature. The closest work is PaperQA [6], an agent that retrieves from full-text scientific articles, judges the relevance of sources and passages, and answers with citations; it outperforms existing language models and agents on science question-answering benchmarks and matches expert researchers on LitQA, a benchmark it introduced. PaperQA2 [12] extends this to match or exceed subject-matter experts on retrieval, summarisation and contradiction detection, evaluated on the harder LitQA2. These systems search broadly across a large literature. Our setting is narrower and more controlled, with three differences. We work on a closed corpus built from the citation graph around a few seed papers, where every answer can be traced to a known passage. Our benchmark records evidence at the level of a quoted sentence, so retrieval can be scored separately from answering. And we compare an agent with a single-shot baseline that shares its prompt rules, retriever and models, so that the difference measures the agent's search strategy alone.

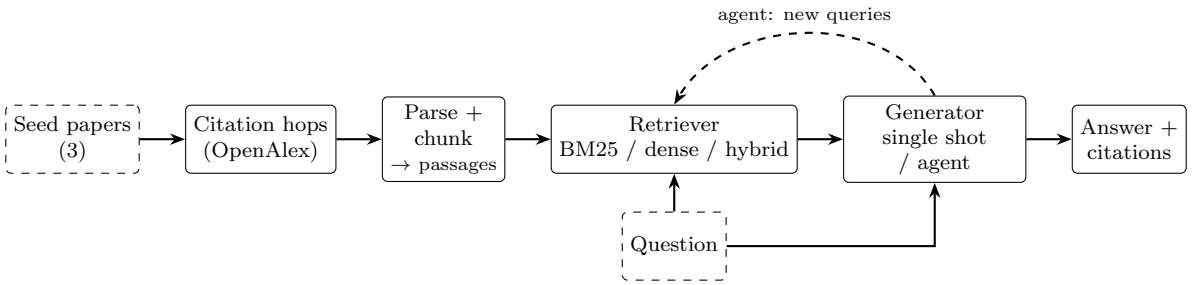


Figure 1: Pipeline. The agent can query the retriever repeatedly (dashed arrow); the single-shot generator queries it once, with the question.

Lexical, dense and hybrid retrieval. Lexical retrieval with BM25 [10] matches query terms against documents and remains a strong baseline, especially for rare technical terms and names. Dense retrieval embeds queries and passages with a neural encoder, such as the BGE family [14], and matches them by meaning, which helps with paraphrase but can blur closely related terms. The two make different errors, and fusing their rankings, for example with reciprocal rank fusion [4], often outperforms either alone. Scientific text is a demanding case for both: its vocabulary is specialised, and general-purpose embedding models are not tuned to it.

Evaluating retrieval and answers. Information-retrieval benchmarks rarely have complete relevance labels. The standard remedy is pooling: the top results of several systems are judged, and unjudged passages are treated with care [**?**]. We follow this practice to label passages that answer a question as well as the cited source (Section 4). Grading free-text answers needs judgment, and large language models are increasingly used as judges; they agree well with human preferences on many tasks but have known biases [16]. We therefore report the agreement between our judge and our own grades (Cohen's κ [3]) next to every judged score.

3 Methodology

The system answers questions about a fixed corpus of papers and cites, for every claim, the passage it comes from. It is built as a pipeline of separable stages (Figure 1):

- a corpus is collected from the citation graph around a few seed papers
- the papers are parsed and split into passages
- a retriever ranks the passages for a query
- a generator writes the answer from the passages it has read

We compare two generators that share everything except how they gather evidence: a *single-shot* generator that retrieves once with the question, and an *agent* that decides itself what to search and read before answering.

3.1 Corpus construction

This method is extensible on any field of corpus built on academic literature, but for our evaluation purposes, the corpus is built around three seed papers that span the field from its origins to current practice: the high-dimensional neural network potential of Behler and Parrinello [2], the E(3)-equivariant NequIP [1], and a recent review of biomolecular force fields [8]. Starting from the seeds, we take one backward hop (every paper a seed cites) and one forward hop (the 50 most-cited papers that cite each seed) through the OpenAlex graph [9]. The forward hop is capped because the seeds are cited thousands of times; ranking by citation

count favours established follow-up work over recent, less-tested results. Preprint and journal versions of the same paper are merged by normalised title.

A citation neighbourhood, unlike a keyword query, yields a corpus shaped like the literature a researcher actually reads: the work a method builds on and the work that builds on it. It is also small enough that every evaluation result can be inspected by hand, while containing many papers that restate one another's results, which makes retrieval non-trivial.

Full texts are taken only from open-access sources (arXiv first, then other open locations OpenAlex lists), and a download is kept only if it is actually a PDF. For papers without a free full text we keep the abstract, from OpenAlex or, failing that, Semantic Scholar. The resulting corpus has 313 papers: 136 with full text, 159 with only an abstract and 18 with metadata only. PDFs and abstracts can be added by hand and are recorded separately and re-applied whenever the corpus is rebuilt. In the runs presented below the E(3)-equivariant NequIP paper was added by hand.

3.2 Parsing and chunking

PDFs are parsed with PyMuPDF [1] and cleaned with heuristics for scientific layouts: running headers and footers are removed, headings are detected and turned into section paths (for example "2 Methods > 2.1 Equivariant features"), reference lists and publisher boilerplate are dropped, and equations split across lines are merged. Mathematics is the known weakness of this parser: it keeps inline symbols but often garbles display equations. We therefore tag questions whose answer is an equation and report them separately (Section 4), rather than switching to a heavier parser before its benefit is measured.

Documents are split into passages ("chunks") of about 400 words, packed paragraph by paragraph, with the last sentences (up to 60 words) of each chunk repeated at the start of the next. The corpus yields [N] chunks.

3.3 Retrieval

We compare three retrievers (sparse, dense and hybrid). All of them index the chunk text together with its title and section.

BM25. A lexical baseline [10] with the usual parameters ($k_1 = 1.2$, $b = 0.75$). The score of chunk d for query q is

$$\mathrm{BM25}(q,d) = \sum_{t \in q} \mathrm{IDF}(t)\, \frac{f(t,d)\,(k_1+1)}{f(t,d) + k_1\left(1 - b + b\,\frac{|d|}{\mathrm{avgdl}}\right)}, \tag{1}$$

where $f(t,d)$ is the frequency of term t in d and $|d|/\mathrm{avgdl}$ its length relative to the average chunk. The tokenizer is written for this domain: names such as `TIP3P`, `ff14SB`, `E(3)` or `DFT-D3` stay whole tokens, and hyphenated terms are indexed both whole and by part, so that "E(3)-equivariant" matches queries for either word.

Dense. Each chunk and each query is embedded with BGE-small-en-v1.5 [14], a 33M-parameter encoder producing 384-dimensional vectors, and chunks are ranked by cosine similarity. Queries are prefixed with the instruction the model was trained with ("Represent this sentence for searching relevant passages:"); passages are not. The model reads at most 512 tokens, so the end of the longest chunks is not embedded[: N of them]. We chose a small model that runs on a CPU; the corpus is embedded once (about a minute) and cached. With a few thousand chunks the index is a single matrix and a search is one matrix–vector product, so no vector database is needed.

Hybrid. BM25 and dense rankings are merged by reciprocal rank fusion [4]. Each retriever returns its top 100 chunks, and chunk d receives

$$\mathrm{RRF}(d) = \sum_{r \in \{\mathrm{BM25},\,\mathrm{dense}\}} \frac{1}{k + \mathrm{rank}_r(d)}, \qquad k = 60. \tag{2}$$

[1] https://pypi.org/project/pymupdf/

Tool	Returns
`search`(query, k, paper, year)	the top k passages for a query (k defaults to 5, at most 10), optionally within one paper or from a given year on
`read_around`(passage, before, after)	the neighbouring passages of the same paper, in reading order
`open_paper`(paper)	title, authors, year, abstract and the list of sections with the first passage of each
`give_up`(reason)	ends the loop without an answer; the reason may cite what the papers do cover

Table 1: The agent's tools. Each is a thin wrapper over the same corpus interface the evaluation uses.

RRF uses only ranks, so BM25 scores and cosine similarities never need to be put on one scale, and it has no weight to tune. The motivation is complementary failure modes: BM25 matches exact technical terms but misses paraphrase ("rotational symmetry" vs. "equivariant"), whereas dense retrieval captures paraphrase but blurs closely related names.

3.4 Answer generation

Both generators follow the same rules, stated in their prompts: use only information in the passages read, cite every claim with the identifier of its passage in square brackets, decline rather than guess, and answer in a few sentences of plain prose. Keeping the rules identical means that differences between them measure how evidence is gathered, not prompt style. After generation, every cited identifier is checked against the passages the generator was actually given; an identifier it never saw is counted as an invalid citation rather than trusted. Prompts are versioned and the version is stored with each result.

Single shot. The question is used as the only query; the top $k = 5$ passages are given to the model, which returns a JSON object with the answer and a flag `answerable`. The flag is set to false whenever the passages do not answer the question, even if some are related.

Extractive baseline. As a floor that needs no language model, we return the sentence from the top k passages with the largest word overlap with the question, with its citation, or decline when no sentence overlaps by at least two content words.

3.4.1 Agent

The agent replaces the single retrieval step with a loop in which the model chooses its own actions, in the style of ReAct [15] and PaperQA [6], using the native tool-use interface of the Claude API. It receives the question, a system prompt describing the tools and the answering rules, and the tools of Table 1. In each turn it either calls one or more tools, whose results are appended to the conversation, or replies without a tool call; that reply is the final answer (Algorithm 1).

Termination. Two budgets bound the cost: at most 8 model calls and 10^5 tokens per question. When either is reached, the next turn carries an instruction to answer now; tool calls made after that are not executed, and the model is told so once before the loop stops. Declining is an explicit action, `give_up`, so that it can be scored. The prompt states that a plain reply always counts as an answer and that `give_up` is the only way to decline, also when related passages were found.

Implementation Implementation details can be found in Appendix A.

Algorithm 1: Agent loop. check validates citations against S, the passages actually read.

input : question q, tools T, turn budget $N = 8$, token budget $B = 10^5$
output: answer with citations, or a decline

$m \leftarrow [q]$; $S \leftarrow \emptyset$; // conversation, passages seen
for $t = 1, 2, \ldots$ **do**
  if $t \geq N$ ***or*** *tokens used* $\geq B$ **then** append "answer now" to m;
  $r \leftarrow \text{LLM}(\text{system}, m, T)$; // same T in every turn
  if r *has no tool call* **then return** check(r.text, S);
  if r *calls* `give_up` **then return** decline(r.reason);
  if *budget exhausted* **then**
    reply that the tools were not run ; // once; then stop
  else
    foreach *tool call* c *in* r **do**
      run c; add returned passages to S; append the result to m

4 Evaluation

4.1 Benchmark

The question set consists of 32 questions, which can be found in Appendix B.

4.2 Retrieval

TODO: BM25, dense (BGE small), hybrid. Report Hit@5 and MRR, strict and lenient; 5 matches the k given to

4.3 End-to-end answers

- 2 x 2 design: Sonnet x single shot, agent x BM25, Hybrid, plus the extractive baseline as a floor.
- Metrics: correct (judge vs. reference), supported (judge, cited passages only), cites evidence,
- Evidence retrieved, invalid citations, declines on unanswerable / answerable; effort and cost.
- Judge: Sonnet 5.5 for every run; validated against your own grades on M answers (agreement, Cohen's kappa [3])

4.4 Statistics

5 Results and discussion

5.1 Retrieval

With the five passages single shot reads, BM25 and hybrid retrieval find the labelled evidence for the same share of answerable questions, 0.74 (37 of 50; Table 2, "evidence retrieved"). They do not find it for the same questions: each finds it on four questions where the other misses it. Hybrid retrieval nonetheless changes how single shot behaves (Section 5.2): it rarely declines. A plausible reason is that, even when the labelled passage is missing, the dense half of the ranking brings in a related passage that lets the model attempt an answer. The labelled evidence measures one kind of success; the strict labels miss passages that restate a result elsewhere, which we return to in Section 5.3.

	Single shot		Agent	
	BM25	Hybrid	BM25	Hybrid
Correct	0.74 [0.63, 0.84]	0.78 [0.69, 0.87]	0.94 [0.89, 0.98]	**0.98** [0.95, 1.00]
Supported†	0.98 (n=42)	0.97 (n=48)	0.95 (n=50)	0.93 (n=50)
Cites evidence†	0.79	0.73	0.84	0.82
Evidence retrieved	0.74	0.74	0.86	0.84
Wrongly declined	0.16	0.04	0.00	0.00
Declined unanswerable (5)	5/5	5/5	4/5‡	5/5
Invalid citations	0	0	0	0
Correct, by question type				
Lookup (37)	0.81	0.84	0.96	0.97
Comparison (9)	0.61	0.56	0.83	1.00
Multi-hop (4)	0.38	0.75	1.00	1.00
Correct, by whether the labelled evidence was among the passages seen				
Found	0.86 (n=37)	0.91 (n=37)	0.95 (n=43)	1.00 (n=42)
Missed	0.38 (n=13)	0.42 (n=13)	0.86 (n=7)	0.88 (n=8)

Table 2: All four configurations on the 55 dev questions (50 answerable, 5 unanswerable; Claude Sonnet 5.5, $k = 5$, judged by Claude Sonnet 5.5). Brackets are 95% bootstrap intervals. †Over answered questions only. ‡The one miss declined in prose instead of calling `give_up` (Section 5.4).

	Correct			Wrongly declined		
Comparison (B vs. A)	Δ [95% CI]	B/W	p	Δ [95% CI]	B/W	p
Agent vs. single shot, BM25	+0.20 [+0.09, +0.31]	14/2	0.004	−0.16 [−0.26, −0.06]	8/0	0.008
Agent vs. single shot, hybrid	+0.20 [+0.12, +0.29]	16/0	<0.001	−0.04 [−0.10, 0.00]	2/0	0.500
Hybrid vs. BM25, single shot	+0.04 [−0.05, +0.14]	8/5	0.581	−0.12 [−0.22, −0.04]	6/0	0.031
Hybrid vs. BM25, agent	+0.04 [0.00, +0.09]	5/1	0.219	0.00 [0.00, 0.00]	0/0	1.000
Agent + hybrid vs. single shot + BM25	+0.24 [+0.14, +0.35]	16/0	<0.001	−0.16 [−0.26, −0.06]	8/0	0.008

Table 3: Paired comparisons on the same 50 answerable dev questions. Δ is the mean per-question difference B − A with a paired bootstrap 95% interval; B/W counts the questions on which B did better / worse than A (for wrongly declined, "better" means it declined less); p is a two-sided exact sign test on those counts. Correctness is graded by Claude Sonnet 5.5 against the reference answers (correct 1, partial 0.5, incorrect 0).

5.2 End-to-end answers

Table 2 and Figure 3 show the four configurations. The agent is the larger effect. It raises correctness from 0.74 to 0.94 with BM25 and from 0.78 to 0.98 with hybrid retrieval, a gain of 0.20 with either retriever (Table 6). With hybrid retrieval it does better on 16 questions and worse on none ($p < 0.001$); with BM25, better on 14 and worse on 2 ($p = 0.004$). The retriever is the smaller effect: hybrid adds 0.04 to correctness for both single shot and the agent, and neither difference is distinguishable from zero on 50 questions. The full system, the agent with hybrid retrieval, answers 48 of the 50 answerable questions fully correctly and the other two partially, against 32 fully correct for single shot with BM25.

The retriever matters more for how often single shot declines. With BM25, single shot declines 8 of the 50 questions the corpus answers (0.16). Hybrid retrieval cuts this to 2 (−0.12, 6 better and 0 worse, $p = 0.031$), and the agent declines none with either retriever. All five unanswerable questions are declined by every configuration except the agent with BM25, whose one miss is a decline that was not recorded as one (Section 5.4).

Faithfulness to the cited passages is high everywhere: between 0.93 and 0.98 of answered questions are judged supported, and no configuration produced an invalid citation. The agent is slightly lower (0.95 and 0.93 against 0.98 and 0.97 for single shot), but the differences are within noise (with hybrid retrieval, −0.04 [−0.10, +0.01], 1 better and 4 worse, $p = 0.375$). The agent answers more questions, including the

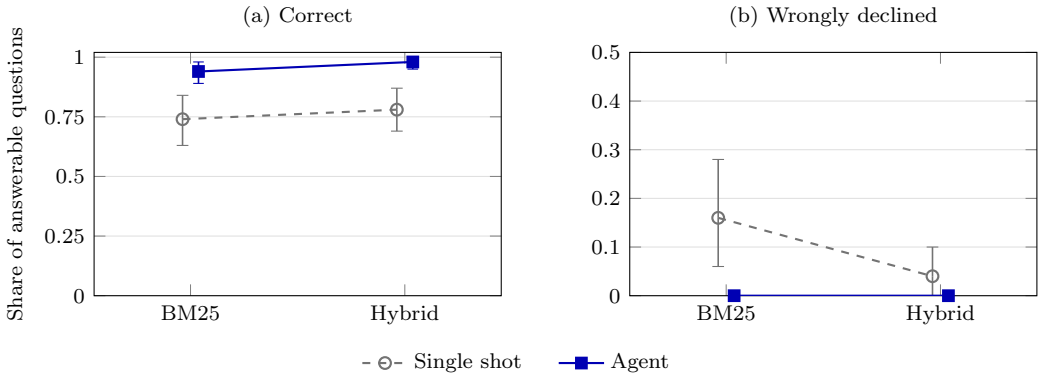


Figure 2: Answer quality for single shot and the agent with BM25 and hybrid retrieval (Claude Sonnet 5.5, 50 answerable dev questions, judged by Claude Sonnet 5.5). Points are means, bars 95% bootstrap intervals of each run. (a) The agent raises correctness by the same 0.20 with either retriever; hybrid retrieval adds a further 0.04 to each. (b) Single shot with BM25 declines 16% of the questions the corpus answers, when its one search misses the evidence; hybrid retrieval removes most of these, and the agent declines none. Paired comparisons are in Table 6.

comparisons and multi-hop questions single shot declined, and writes longer answers, so it makes more claims that each need a passage. In almost every partially supported answer, of any configuration, a single claim lacks support, and it is usually a linking fact the model supplied from its own knowledge: that GDML is a kernel method, that DimeNet++ builds on DimeNet, or who wrote a cited paper.

By question type (Table 2), the agent's gain is largest where one search is least likely to be enough. On lookups it moves single shot's 0.81–0.84 to 0.96–0.97. On comparisons, which need passages from two papers, it moves 0.56–0.61 to 0.83 with BM25 and 1.00 with hybrid. On multi-hop questions, where the second search depends on the first answer, single shot with BM25 scores 0.38 and the agent 1.00. There are only nine comparison and four multi-hop questions, so these per-type numbers are indicative.

5.3 Where the gain comes from

Single shot's errors are mostly retrieval errors. With BM25 it is less than fully correct on 18 questions, and on 11 of them the labelled evidence was not among its five passages; with hybrid retrieval it is 12 of 18. When the evidence is retrieved, single shot is correct 0.86–0.91 of the time; when it is not, only 0.38–0.42 (Table 2). How it fails differs by retriever: with BM25 it declines (8 of its 18 errors), with hybrid retrieval it answers partially from the passages it has (2 declines of 18).

The agent repairs these by searching again. It searched once on 35 of the 55 questions and more than once on the other 20, and the extra searches go where they are needed: on average 1.1–1.2 searches for lookups, 1.9–2.1 for comparisons and 1.8–2.0 for multi-hop questions. Of the 14 questions it improves with BM25, 8 are questions single shot wrongly declined and 7 are questions where single shot's search missed the evidence; with hybrid retrieval, 10 of its 16 improvements are missed-evidence questions. The other tools were almost unused: `read_around` was called twice in 55 questions with hybrid retrieval and never with BM25, and `open_paper` never. The gain therefore comes from reformulating and repeating searches, not from reading passages in context.

The agent also answers correctly where it never saw the labelled passage. Of the 13 questions on which single shot with BM25 missed the evidence, the agent found the labelled passage for 9 but answered 11 correctly; with hybrid retrieval it found 6 of 13 and answered 12 correctly. It cites passages that answer the question but are not the ones we labelled, often a later paper restating a result or a second version of the same paper (Section 5.4). Strict evidence labels therefore understate retrieval, and "evidence retrieved" and "cites evidence" should be read as lower bounds; the lenient labels from pooled review (Section 4) address this.

5.4 Error analysis

Four traces illustrate the main patterns.

A multi-hop question the agent fixes. Asked which empirical potential DiffTRe's diamond model starts from, and which neural network potential paper used that potential's element as its test system, single shot with BM25 retrieves passages about carbon allotropes and about Behler–Parrinello on silicon, but nothing from DiffTRe, and declines. The agent first searches for the DiffTRe diamond model, learns that it uses the Stillinger–Weber potential for silicon rescaled for carbon, then searches for neural network potentials tested on silicon, and finds the Behler–Parrinello paper. The second query depends on the first answer, which a single search cannot provide. The support judge rated this answer only partially supported, because the authors' names come from the paper's metadata rather than from the cited passage text: a limitation of judging support on passage text alone.

A decline that was not recorded. Asked for AlphaFold's learning rate and optimizer, which the paper defers to its supplementary material, the agent with BM25 searched five times, including within the AlphaFold paper, and then replied that the passages do not give either value, quoting what the paper does say. This is the correct behaviour, but the agent replied in prose instead of calling `give_up`, so it was scored as an answer to an unanswerable question. With hybrid retrieval the same question was declined through `give_up`. Even with the prompt stating that a plain reply always counts as an answer, the model occasionally declines in prose; the abstention metrics should be read with this in mind.

The question, not the system. Asked how SchNet turns an interatomic distance into the input of its filter network, both agents described the Gaussian expansion correctly but were graded partial for its parameters (a 0–30 Å grid and $\gamma = 10$ Å, where the reference gives 0.1 Å). Both took them, with a correct citation, from the earlier conference paper on SchNet, which is also in the corpus, while the question's evidence is the later journal paper. The answer is faithful to its source; the question does not say which paper it means. Questions about a method described in several papers need to name the paper.

Equations. Across the six questions tagged *equation*, the agent with hybrid retrieval is 0.92 correct but only 0.75 supported, and it cites the labelled evidence on 2 of 6. Asked how DiffTRe computes its reweighting weights, it gave the correct formula and the reason the kinetic energy cancels, graded correct, but cited the passage after the one that states the cancellation, and the support judge found neither the formula nor the cancellation in what it cited. Asked for the cutoff function of the original Behler–Parrinello potential, it gave the correct cosine form, but the judge noted that the equation in the cited passage is garbled, so π and the sign in the formula cannot be read there. The model can know an equation the passage text no longer shows; equations are where parsing quality limits citation quality most (Section 3.2).

5.5 Cost

Table 4 gives effort and cost. The agent makes 2.4 model calls per question on average and reads 7.0–7.5 passages, against one call and five passages for single shot. Its input tokens per question are nearly three times higher, but a third of them are cache reads of the conversation so far, billed at 5% of the input price, so the cost per question only doubles: about USD 0.010 against 0.005. For the full dev run of 55 questions that is USD 0.55–0.57 against 0.27–0.28. The 0.20 gain in correctness thus costs about half a cent per question. The budgets were never the limit: no run hit the eight-call or 100,000-token budget, and the longest trace used seven calls.

6 Limitations

7 Conclusion and future work

Appendix

	Single shot		Agent	
Per question (mean)	BM25	Hybrid	BM25	Hybrid
Model calls	1.00	1.00	2.36	2.35
Searches	1.00	1.00	1.67	1.51
Passages read	5.0	5.0	7.5	7.0
Input tokens	4,200	3,986	12,779	11,355
of which cache reads	0	0	5,110	4,267
Output tokens	275	276	449	450
Cost (USD)	0.0051	0.0049	0.0103	0.0099
Whole run, generator (USD)	0.28	0.27	0.57	0.55

Table 4: Effort and cost of answering, per question and for the 55-question dev run. The judge, which grades every configuration the same way, is not included. The agent costs about twice as much per question; a third of its input tokens are cache reads of the conversation so far.

A Implementation

The system is a Python library with a command-line interface for building the corpus, running the evaluations and asking questions, and a small web application (FastAPI and React) on top of the same functions for writing questions, reviewing retrieved passages and inspecting results and traces. The models are Claude Haiku 5.5 and Claude Sonnet 5.5, accessed through the Anthropic API; the dense retriever runs locally with sentence-transformers. All components, including the agent loop, are tested offline with a scripted model that stands in for the language model.

Cost. Because every turn resends the conversation, we enable prompt caching, which bills the repeated prefix at 5–10% of the input price, and cache every request and response on disk, which makes repeated evaluations free and reproducible. Every answer records each step (tool, arguments, returned passages, tokens and time), which serves both error analysis and the cost comparison in Section D.

Design choices. Four details proved necessary in practice. (i) The list of tools is identical in every turn. Current models return signed reasoning ("thinking") blocks that must be passed back unchanged, and the signature is bound to the tool list; an earlier version that offered only `give_up` on the last turn was rejected by the API for this reason. (ii) A passage is shown to the model in full only the first time it is returned; later results repeat only its identifier and heading, which keeps the conversation, and its cost, from growing with repeated hits. (iii) Each search returns the same k as single shot by default, and the tool description states that default, so the agent starts from the same evidence; the model may request up to 10. (iv) The final answer is a reply without a tool call rather than a dedicated `answer` tool, because in early runs the smaller model ignored such a tool and answered in text.

B Question set

The benchmark has 55 questions (37 lookup, 9 comparison, 4 multi-hop, 5 unanswerable). Each answerable question has a reference answer and its evidence: the paper and a short quote, which the evaluation matches to passages by word-trigram overlap. Quotes are shown as extracted by the PDF parser, so equations appear garbled. Alternatives are passages from other papers or places that answer the question too; they count for lenient scoring.

Lookup questions (37)

One fact from one passage.

7 **q001** LOOKUP · `behler-parrinello`

Question. What test system did Behler and Parrinello use to demonstrate their neural network potential?

Type	Questions	Tag	Questions
Lookup	37	`alphafold`	15
Comparison	9	`numeric`	14
Multi-hop	4	`esmfold`	13
Unanswerable	5	`difftre`	12
		`definition`	10
		`schnet`	10
		`nequip`	8
		`behler-parrinello`	6
		`equation`	6
		`out-of-scope`	1
		`force-fields`	1

Table 5: Question types and tags. A question can have several tags; `equation` marks answers that are formulas.

Reference answer. Bulk silicon; the network potential was compared against DFT reference calculations.
Evidence. Behler & Parrinello [2]: "The high accuracy of the method is demonstrated for bulk silicon and compared with empirical potentials and DFT"
Judgments: 1 alternative passage.

7 q002 LOOKUP · `equation`, `behler-parrinello`
Question. What is the functional form of the cutoff function in the original high-dimensional neural network potential?
Reference answer. A cosine cutoff: $f_c(R) = \frac{1}{2}\left[\cos(\pi R/R_c) + 1\right]$ for $R \leq R_c$, and 0 beyond the cutoff radius R_c.
Evidence. Behler & Parrinello [2]: "In order to define the energetically relevant local environment we employ a cutoff function fc of the interatomic distance Rij, which has the form fcRij 8< : 0:5 h cos Rij Rc 1 i for Rij Rc; 0 for Rij > Rc:"
Judgments: 3 judged not relevant.

7 q003 LOOKUP · `equation`, `definition`, `behler-parrinello`
Question. How are the total energy and the atomic contributions related in a high-dimensional neural network potential?
Reference answer. The total energy is a sum of atomic energies, $E = \sum_i E_i$, each predicted by an atomic neural network from that atom's local environment within a cutoff.
Evidence. Behler & Parrinello [2]: "The main idea is to represent the total energy E of the system as a sum of atomic contributions Ei, an approach that is typically also used in empirical potentials E X i Ei:"

7 q004 LOOKUP · `definition`, `nequip`
Question. Under which symmetry group are NequIP's internal features equivariant, and how are they built?
Reference answer. E(3); features are geometric tensors indexed by irreducible representations of O(3), combined by tensor products with spherical harmonics of relative position vectors.
Evidence. Batzner et al. (NequIP) [1]: "Formally, the feature vectors are geometric objects that comprise a direct sum of irreducible representations of the O(3) symmetry group."
Judgments: 2 alternative passages, 3 judged not relevant.

7 q008 LOOKUP · `nequip`
Question. What does Nequip do as a first step in the architecture?
Reference answer. It embeds each atom's chemical species. The atomic number Z_i, one-hot encoded, goes through a trainable self-interaction (linear) layer. This gives the initial per-atom features, which are scalar ($l = 0$) only. The embedding follows SchNet. Higher-order tensor features first appear in the interaction blocks.
Evidence. Batzner et al. (NequIP) [1]: "Embedding: following SchNet, the initial feature is generated using a trainable embedding that operates on the atomic number Zi (represented via a one-hot encoding) alone, implemented via a trainable self-interaction layer."

7 q009 LOOKUP

Question. What datasets is Nequip tested on?

Reference answer. (1) MD-17 small organic molecules, both the original DFT version and the revised MD-17 (rMD17); (2) small molecules at quantum-chemical accuracy: aspirin at CCSD, and benzene, malonaldehyde, toluene and ethanol at CCSD(T); (3) periodic systems: liquid water and ice, a chemical reaction at a solid/gas interface (formate decomposition on a copper surface), amorphous lithium phosphate ($Li_4P_2O_7$), and a lithium superionic conductor ($Li_{6.75}P_3S_{11}$, "LiPS").

Evidence. Batzner et al. (NequIP) [1]: "We validate the proposed method on a diverse series of challenging data sets: first we demonstrate that we improve upon state-of-the-art accuracy on MD-17, a data set of small, organic molecules that is widely used for benchmarking ML-IPs [9, 11, 35–37]. Next, we show that . . . "

7 q010 LOOKUP

Question. Which other models is Nequip compared to in the original Nequip paper on MD-17?

Reference answer. Invariant GNN potentials: SchNet and DimeNet. A shallow neural network: ANI. Kernel methods: GAP, FCHL19/GPR and sGDML. Other methods: ACE, SpookyNet and GemNet. Other equivariant networks: PaiNN, NewtonNet and UNiTE. The paper notes that PaiNN and NewtonNet are restricted to $l_{\max} = 1$. Results for ACE, GAP and ANI on revised MD-17 are taken from another paper (Kovács et al.), not retrained by the authors. NequIP reports the best accuracy of all of them.

Evidence. Batzner et al. (NequIP) [1]: "We find that NequIP significantly outperforms invariant GNN-IPs (such as SchNet [9] and DimeNet [11]), shallow neural networks (such as ANI [40]), and kernel-based approaches (such as GAP [5], FCHL19/GPR [39, 41] and sGDML [37]). Finally, we compare to a series of other methods including . . . "

7 q011 LOOKUP · `equation`, `schnet`

Question. How does SchNet turn an interatomic distance into the input of its filter network?

Reference answer. It expands the distance in a basis of Gaussians, $e_k = \exp\big(-\gamma\,(\|\mathbf{r}_j - \mathbf{r}_i\| - \mu_k)^2\big)$, with centers μ_k on a uniform grid from zero to the distance cutoff. Grid spacing and γ are both set to 0.1 Å. The expansion decorrelates the filter values, which improves the conditioning of the optimization.

Evidence. Schütt et al. (SchNet) [11]: "with centers μk chosen on a uniform grid between zero and the distance cutoff. This has the effect of decorrelating the filter values which improves the conditioning of the optimization problem."

7 q012 LOOKUP · `equation`, `schnet`

Question. Which activation function does SchNet use, and why does that matter for force fields?

Reference answer. A shifted softplus, $\mathrm{ssp}(x) = \ln(\frac{1}{2}e^x + \frac{1}{2})$. The shift makes $\mathrm{ssp}(0) = 0$ and improves convergence, and because it is infinitely differentiable the learned energy surface is smooth, giving smooth forces and the second derivatives needed for training on forces and for vibrational modes.

Evidence. Schütt et al. (SchNet) [11]: "The shifting ensures that ssp(0) = 0 and improves the convergence of the network while having infinite order of continuity."

7 q013 LOOKUP · `definition`, `schnet`

Question. Why can't SchNet use ordinary convolutional layers like image models do?

Reference answer. Atoms sit at arbitrary positions rather than on a regular grid like pixels, so a fixed filter tensor can't be used. SchNet uses continuous-filter convolutions, where a filter-generating network computes the filter values from the atoms' relative positions.

Evidence. Schütt et al. (SchNet) [11]: "This generalization is necessary since atoms are not located on a regular grid like image pixels, but can be located at arbitrary positions."

7 q014 LOOKUP · `numeric`, `schnet`

Question. What cutoff radius did SchNet use when learning formation energies of bulk crystals?

Reference answer. 5 Å, on the Materials Project data. Two structures that would contain isolated atoms at that cutoff were dropped. (For the small QM9 and MD17 molecules no cutoff was used.)

Evidence. Schütt et al. (SchNet) [11]: "We set the distance cutoff rcut = 5Å and discard two examples from the data set that would include isolated atoms with this setting."

7 q015 LOOKUP · `equation`, `schnet`

Question. How does SchNet make sure its predicted forces conserve energy?

Reference answer. Forces are not predicted directly: they are the negative gradient of the predicted energy with respect to the atomic positions, $\mathbf{F}_i = -\partial E/\partial \mathbf{r}_i$. With a rotationally invariant energy model this gives

rotationally equivariant forces and an energy-conserving force field.
Evidence. Schütt et al. (SchNet) [11]: "When using a rotationally invariant energy model, this ensures rotationally equivariant force predictions and guarantees an energy conserving force field"

7 q023 LOOKUP · `definition`, `difftre`

Question. What does DiffTRe do differently from backpropagating through an MD simulation, and what does that buy it?
Reference answer. It doesn't differentiate through the simulation at all. It combines automatic differentiation with MD reweighting, so gradients flow through the reweighted ensemble averages instead of the trajectory. For time-independent observables this avoids exploding gradients and makes gradient computation about two orders of magnitude faster, with memory needs similar to the adjoint method.
Evidence. Thaler & Zavadlav (DiffTRe) [13]: "DiffTRe offers end-to-end gradient computation and circumvents the need to differentiate through the simulation by combining AD with previous work on MD reweighting schemes"

7 q024 LOOKUP · `equation`, `difftre`

Question. How are the reweighting weights computed in DiffTRe, and why does the kinetic energy drop out?
Reference answer. In the canonical ensemble, each saved state gets $w_i \propto \exp(-\beta\,[U_\theta(S_i) - U_{\mathrm{ref}}(S_i)])$, normalized over all states. Only the potential energy appears because the kinetic energy is the same under the old and new potential and cancels in the Boltzmann ratio. With θ equal to the reference, all weights are $1/N$.
Evidence. Thaler & Zavadlav (DiffTRe) [13]: "Inserting pθ(Si) into Eq. (3) allows computing weights as a function of θ (the kinetic energy cancels)"

7 q025 LOOKUP · `difftre`

Question. When does DiffTRe stop reusing the old trajectory and simulate a new one?
Reference answer. When the effective sample size of the reweighted states, $N_{\mathrm{eff}} \approx \exp(-\sum_i w_i \ln w_i)$, falls below a chosen threshold, meaning a few states dominate the average and its statistical error is too large. A new reference trajectory is then run with the current parameters, starting from the last state of the old one.
Evidence. Thaler & Zavadlav (DiffTRe) [13]: "As soon as Neff $<$ $\bar{N}$eff, a new reference trajectory needs to be sampled using the current θ as the new $\hat{\theta}$."

7 q026 LOOKUP · `numeric`, `difftre`

Question. Which experimental data was the DiffTRe diamond potential trained on?
Reference answer. The experimental elastic stiffness tensor of diamond, which by cubic symmetry has three moduli: $C_{11} = 1079\,\mathrm{GPa}$, $C_{12} = 124\,\mathrm{GPa}$ and $C_{44} = 578\,\mathrm{GPa}$. Training also targeted a stress-free state at zero strain.
Evidence. Thaler & Zavadlav (DiffTRe) [13]: "we learn a DimeNet++ [12] potential for diamond from its experimental elastic stiffness tensor C"

7 q027 LOOKUP · `numeric`, `difftre`

Question. Which held-out property did DiffTRe's coarse-grained water model get wrong, and why?
Reference answer. The self-diffusion coefficient: about 10.9 versus $2.2\,\mu\mathrm{m}^2/\mathrm{ms}$ in experiment, so diffusion is too fast. Coarse-grained models have smoother energy surfaces, which speeds up their dynamics compared with atomistic models. (The tetrahedral order parameter, by contrast, matched closely.)
Evidence (one passage per part). Thaler & Zavadlav (DiffTRe) [13]: "The learned CG water model predicts a larger self-diffusion coefficient than were experimentally measured"; Thaler & Zavadlav (DiffTRe) [13]: "Due to smoother PESs, CG models exhibit accelerated dynamical processes compared to atomistic models"

7 q028 LOOKUP · `difftre`

Question. Could you use DiffTRe to fit a potential to a measured diffusion coefficient?
Reference answer. No. Reweighting treats the trajectory as a set of states and ignores their order in time, so dynamical properties such as diffusion coefficients can't be targets; only time-independent observables can.
Evidence. Thaler & Zavadlav (DiffTRe) [13]: "The reweighting scheme renders DiffTRe invariant to the sequence of states in the trajectory. Hence, dynamical properties cannot be employed as target observables."

7 q032 LOOKUP · `definition`, `difftre`

Question. In DiffTRe, what is the prior potential for, if not a Bayesian prior?

Reference answer. A simple physics-based potential (e.g. a repulsive Lennard-Jones term, or Stillinger-Weber) added to the neural network. Early in training it controls which configurations the simulations visit, and therefore where the gradient carries information. It also gives a physics-informed starting point, so the network learns a correction to the prior. If the data contradicts the prior, the network overrides it.
Evidence. Thaler & Zavadlav (DiffTRe) [13]: "Rather, the prior serves to control the data (trajectory) generation in the beginning of the optimization."

7 q034 LOOKUP · `numeric`, `alphafold`

Question. How close to the experimental structures were AlphaFold's backbone predictions at CASP14, and how did the runner-up do?
Reference answer. Median backbone accuracy of 0.96 Å r.m.s.d.95 (Cα r.m.s.d. at 95% residue coverage; 95% CI 0.85–1.16 Å), against 2.8 Å for the next best method. For scale, a carbon atom is about 1.4 Å wide.
Evidence. Jumper et al. (AlphaFold) [**?**]: "AlphaFold structures had a median backbone accuracy of 0.96 Å r.m.s.d.95 (Cα root-mean-square deviation at 95% residue coverage)"

7 q035 LOOKUP · `definition`, `alphafold`

Question. What are the two main stages of the AlphaFold network, and what does each one produce?
Reference answer. First a trunk of repeated Evoformer blocks (48) turns the inputs into an Nseq × Nres MSA representation and an Nres × Nres pair representation. Then the structure module (8 blocks, shared weights) builds an explicit 3D structure as a rotation and translation for each residue, starting from all residues at the origin with identity rotations.
Evidence (one passage per part). Jumper et al. (AlphaFold) [**?**]: "First, the trunk of the network processes the inputs through repeated layers of a novel neural network block that we term Evoformer"; Jumper et al. (AlphaFold) [**?**]: "The trunk of the network is followed by the structure module that introduces an explicit 3D structure in the form of a rotation and translation for each residue of the protein"

7 q036 LOOKUP · `alphafold`

Question. Why are AlphaFold's pair-representation updates organised around triangles of residues?
Reference answer. For pairwise information to correspond to a single 3D structure it must be consistent, e.g. distances must satisfy the triangle inequality. So each update of an edge ij uses the other two edges of triangles ijk: triangle self-attention with an extra bias for the missing edge, and the triangle multiplicative update. Either alone gives accurate structures, but combining both is more accurate.
Evidence. Jumper et al. (AlphaFold) [**?**]: "many constraints must be satisfied including the triangle inequality on distances. On the basis of this intuition, we arrange the update operations on the pair representation in terms of triangles of edges involving three different nodes"

7 q037 LOOKUP · `alphafold`, `force-fields`

Question. Does AlphaFold's structure module keep the peptide bonds intact while it builds the structure? How is the geometry fixed at the end?
Reference answer. No. Each residue is a free-floating rigid body ("residue gas"), and the network often breaks the chain, which lets it refine all parts locally without solving loop closure. A violation loss encourages correct peptide geometry during fine-tuning, but it is only enforced exactly by a final relaxation by gradient descent in the Amber force field (Amber99sb in OpenMM), which removes stereochemical violations without improving accuracy.
Evidence (one passage per part). Jumper et al. (AlphaFold) [**?**]: "the peptide bond geometry is completely unconstrained and the network is observed to frequently violate the chain constraint during the application of the structure module"; Jumper et al. (AlphaFold) [**?**]: "Exact enforcement of peptide bond geometry is only achieved in the post-prediction relaxation of the structure by gradient descent in the Amber32 force field."

7 q038 LOOKUP · `definition`, `alphafold`

Question. What loss is AlphaFold mainly trained with, and what does it compare?
Reference answer. The frame-aligned point error (FAPE). For each residue's frame, the predicted structure is aligned to the true one on that frame and the distances between predicted and true atom positions are computed; the Nframes × Natoms distances are penalised with a clamped L1 loss. This makes atoms correct relative to each residue's local frame and is the main source of chirality.
Evidence. Jumper et al. (AlphaFold) [**?**]: "The final loss (which we term the frame-aligned point error (FAPE) (Fig. 3f)) compares the predicted atom positions to the true positions under many different alignments."

7 q039 LOOKUP · `alphafold`

Question. How did AlphaFold make use of protein sequences that have no known structure?

Reference answer. Self-distillation, similar to noisy student: a trained network predicted structures for about 350,000 Uniclust30 sequences (355,993 in the Methods), kept a high-confidence subset, and a new model was trained from scratch on a mix of PDB data and these predictions (75% predictions, 25% PDB). A BERT-style masked-MSA objective also uses unlabelled information, trained jointly rather than pre-trained.

Evidence. Jumper et al. (AlphaFold) [**?**]: "we use a trained network to predict the structure of around 350,000 diverse sequences from Uniclust3036 and make a new dataset of predicted structures filtered to a high-confidence subset"

7 q040 LOOKUP · `numeric`, `alphafold`

Question. Below what alignment depth does AlphaFold's accuracy drop off, and what other kind of protein does it struggle with?

Reference answer. Accuracy falls substantially when the median MSA depth is below about 30 sequences; above about 100 the gains are small. It is also much weaker for proteins with few intra-chain or homotypic contacts compared with heterotypic ones, such as bridging domains whose shape comes from other chains in a complex.

Evidence (one passage per part). Jumper et al. (AlphaFold) [**?**]: "the accuracy decreases substantially when the median alignment depth is less than around 30 sequences"; Jumper et al. (AlphaFold) [**?**]: "AlphaFold is much weaker for proteins that have few intra-chain or homotypic contacts compared to the number of heterotypic contacts"

7 q041 LOOKUP · `numeric`, `alphafold`

Question. How long does a single AlphaFold model take for a 384-residue protein on a V100, with and without ensembling?

Reference answer. 9.2 minutes with ensembling (the CASP14 configuration) and 1.1 minutes without; turning ensembling off makes the network about 8× faster at very similar accuracy.

Evidence (one passage per part). Jumper et al. (AlphaFold) [**?**]: "Representative timings for the neural network using a single model on V100 GPU are 4.8 min with 256 residues, 9.2 min with 384 residues and 18 h at 2,500 residues."; Jumper et al. (AlphaFold) [**?**]: "Without ensembling, the network is 8× faster and the representative timings for a single model are 0.6 min with 256 residues, 1.1 min with 384 residues and 2.1 h with 2,500 residues."

7 q042 LOOKUP · `numeric`, `alphafold`

Question. Which ablation shows AlphaFold needs metagenomic sequence databases, and how big was the effect?

Reference answer. Re-running CASP14 with different MSA sources: removing BFD cost 0.4 GDT, removing MGnify 0.7 GDT, and removing both 6.1 GDT. Most targets barely changed, but a few outliers changed by 20+ GDT; at least one metagenomic database matters for targets poorly covered by UniRef.

Evidence. Jumper et al. (AlphaFold) [**?**]: "Removing BFD reduced the mean accuracy by 0.4 GDT, removing Mgnify reduced the mean accuracy by 0.7 GDT, and removing both reduced the mean accuracy by 6.1 GDT."

7 q046 LOOKUP · `definition`, `esmfold`

Question. What does ESMFold use instead of a multiple sequence alignment, and what does that save at prediction time?

Reference answer. The internal representations of the ESM-2 protein language model, computed from the single input sequence. No external evolutionary databases, MSAs or templates are needed, which removes the search for related sequences (over 10 minutes with AlphaFold's and RosettaFold's high-sensitivity pipelines) and simplifies the network.

Evidence. Lin et al. (ESMFold) [**?**]: "This removes costly aspects of current state-of-the-art structure prediction pipelines, eliminating the need for a multiple sequence alignment, while at the same time greatly simplifying the neural architecture used for inference."

7 q047 LOOKUP · `numeric`, `esmfold`

Question. What data was ESM-2 trained on, and how many distinct sequences does it see during training?

Reference answer. UniRef sequences, with masked language modelling: sampled evenly across ˜43 million UniRef50 clusters drawn from ˜138 million UniRef90 sequences, so the model sees ˜65 million unique sequences over training.

Evidence. Lin et al. (ESMFold) [**?**]: "During training sequences are sampled with even weighting across ∼43 million

UniRef50 training clusters from ∼138 million UniRef90 sequences so that over the course of training the model sees ∼65 million unique sequences."

7 q048 LOOKUP · `numeric`, `definition`, `esmfold`

Question. What does perplexity mean for a protein language model, and how much did it improve from the smallest to the largest ESM-2?

Reference answer. Roughly the number of amino acids the model is choosing between at each prediction: 1 for a perfect model, 20 for random guessing. At 270K training steps the 8M-parameter model has a perplexity of 10.45 and the 15B model 6.37.

Evidence (one passage per part). Lin et al. (ESMFold) [**?**]: "This fidelity can be measured using perplexity, which ranges from 1 for a perfect model to 20 for a model that makes predictions at random."; Lin et al. (ESMFold) [**?**]: "At 270K training steps the 8M parameter model has a perplexity of 10.45, and the 15B model reaches a perplexity of 6.37"

7 q049 LOOKUP · `numeric`, `esmfold`

Question. How accurate is ESMFold compared with AlphaFold2 and RoseTTAFold on CAMEO and CASP14?

Reference answer. Average TM-score 0.83 on CAMEO and 0.68 on CASP14, against 0.88 and 0.85 for AlphaFold2 with MSAs and templates. RoseTTAFold averages 0.82 on CAMEO, so ESMFold is competitive with it there. With the MSA removed, AlphaFold and RoseTTAFold fall well below ESMFold.

Evidence. Lin et al. (ESMFold) [**?**]: "ESMFold achieves an average TM-score of 0.83 on CAMEO and 0.68 on CASP14. Using the search protocols released with AlphaFold2, including MSAs and templates, AlphaFold2 achieves 0.88 and 0.85 on CAMEO and CASP14 respectively."

7 q050 LOOKUP · `numeric`, `esmfold`

Question. Which part of ESMFold matters most according to its ablations?

Reference answer. The language model. Against a smaller 8-block baseline at 0.74 LDDT on CAMEO, removing the language model drops single-sequence performance to 0.58 LDDT, and removing the folding trunk drops it to 0.66. Other ablations (one structure-module block, no recycling, no AlphaFold2 distillation targets, no triangular updates) cost only 0.01–0.04 LDDT.

Evidence. Lin et al. (ESMFold) [**?**]: "Without the language model, the single sequence performance on the CAMEO test set degrades substantially, to 0.58 LDDT."

7 q051 LOOKUP · `esmfold`

Question. Which ESM-2 model does ESMFold use, and is it fine-tuned along with the folding head?

Reference answer. The 3B-parameter ESM-2, the largest that allows inference on a single GPU, and its parameters are frozen while ESMFold is trained.

Evidence. Lin et al. (ESMFold) [**?**]: "Language model parameters are frozen for training ESMFold. We use the 3B parameter language model, the largest model that permits inference on a single GPU."

7 q052 LOOKUP · `definition`, `esmfold`

Question. Why does ESM-2 use rotary position embeddings instead of the learned absolute positions of ESM-1b?

Reference answer. Absolute position encodings (sinusoidal or learned) don't extrapolate beyond the context window seen in training; RoPE lets the model extrapolate. It costs a little extra computation and improved small models, but the gains faded as model size and training duration grew.

Evidence. Lin et al. (ESMFold) [**?**]: "In ESM-2, we used Rotary Position Embedding (RoPE) (66) to allow the model extrapolate beyond the context window it is trained on."

7 q053 LOOKUP · `numeric`, `esmfold`

Question. How big is the ESM Metagenomic Atlas, and how long did it take to compute?

Reference answer. Over 617 million MGnify90 proteins (all sequences of length 20–1024, 99% of the database), folded in 2 weeks on about 2,000 GPUs. About 225 million predictions are high confidence (mean pLDDT > 0.7 and pTM > 0.7) and about 365 million have good confidence (> 0.5).

Evidence (one passage per part). Lin et al. (ESMFold) [**?**]: "We were able to complete the predictions in 2 weeks on a cluster of approximately 2,000 GPUs"; Lin et al. (ESMFold) [**?**]: "∼225 million predictions with high confidence (mean pLDDT > 0.7 and pTM > 0.7)"

7 q054 LOOKUP · `numeric`, `esmfold`

Question. How many of the high-confidence metagenomic structures have no similar experimentally determined structure?

Reference answer. In a random sample of 1 million high-confidence structures, 12.6% (125,765) have no PDB structure with TM-score over 0.5; at a 0.7 threshold it is 25.4%. 10.4% have both low structural similarity and no close sequence homolog.

Evidence. Lin et al. (ESMFold) [**?**]: "For 12.6% of the structures (125,765 proteins), no structure is found with TM-score over 0.5 in the PDB database"

Comparison questions (9)

Two passages compared side by side.

7 q005 COMPARISON · `behler-parrinello`, `nequip`

Question. How does NequIP's handling of rotational symmetry differ from the Behler-Parrinello symmetry functions?

Reference answer. Behler-Parrinello uses fixed, hand-designed invariant descriptors (scalars) of the environment; NequIP learns equivariant features that carry vector/tensor information through message passing, and only the final energy is invariant.

Evidence (one passage per part). Behler & Parrinello [2]: "Furthermore, the symmetry function values must be invariant"; Batzner et al. (NequIP) [1]: "ven though the output of NequIP is the predicted potential energy Epot, which is invariant under translations, reflection, and rotations, the network contains internal features that are geometric tensors which are equivariant to rotation and reflection. This constitutes the core difference between NequIP and existing . . . "

Judgments: 1 alternative passage.

7 q007 COMPARISON · `nequip`

Question. What features does the Nequip architecture use, that are also used by schnet?

Reference answer. Several design choices come from SchNet. The initial atom features are a trainable embedding of the atomic number (one-hot encoded). Atoms then pass through a stack of interaction blocks that do continuous convolutions over neighbors within a cutoff. In those convolutions the learnable filter weights are an MLP of the interatomic distance expanded in a radial basis: Gaussians in SchNet, Bessel functions with an envelope in NequIP. The output is per-atom energies summed to a total energy, and forces are its negative gradient. The difference is that NequIP also multiplies the radial part by spherical harmonics of the bond direction. If only $l = 0$ (scalar) features are kept, NequIP reduces to a SchNet-style invariant network.

Evidence (one passage per part). Batzner et al. (NequIP) [1]: "The NequIP network architecture, shown in Figure 2, is built on an atomic embedding, followed by a series of interaction blocks, and finally an output block: • Embedding: following SchNet, the initial feature is generated using a trainable embedding that operates on the atomic number"; Batzner et al. (NequIP) [1]: "making the network a conventional invariant GNN-IP, involving only invariant convolutions over scalar features in a SchNet-style fashion"

7 q016 COMPARISON · `schnet`, `nequip`

Question. SchNet and NequIP both use continuous convolutions over neighboring atoms. What do their filters see about the neighbors, and how does that change the features they can build?

Reference answer. SchNet's filters only see interatomic distances (expanded in Gaussians), so they are rotationally invariant and the features stay scalars. NequIP's filters also use the direction to the neighbor, so its internal features are geometric tensors that are equivariant to rotation; restricted to scalars it reduces to a SchNet-style invariant network.

Evidence (one passage per part). Schütt et al. (SchNet) [11]: "It is straightforward to include rotational invariance by computing pairwise distances instead of using relative positions."; Batzner et al. (NequIP) [1]: "making the network a conventional invariant GNN-IP, involving only invariant convolutions over scalar features in a SchNet-style fashion"

7 q018 COMPARISON · `schnet`, `behler-parrinello`

Question. Both SchNet and the Behler-Parrinello potential build the prediction from atomic contributions. For which properties does the SchNet paper say this is a poor fit?

Reference answer. Behler–Parrinello writes the total energy as a sum of atomic energies. SchNet does the

same for its targets and does noticeably worse on the polarizability and the electronic spatial extent $\langle R^2 \rangle$ of QM9 molecules, which the authors attribute to that atom-wise decomposition not suiting these properties.
Evidence (one passage per part). Behler & Parrinello [2]: "The main idea is to represent the total energy E of the system as a sum of atomic contributions Ei, an approach that is typically also used in empirical potentials"; Schütt et al. (SchNet) [11]: "This is most likely due to the decomposition of the energy into atomic contributions which is not appropriate for these properties."

7 **q029** COMPARISON · `difftre`, `schnet`

Question. SchNet's C20 force field was trained on DFT data. What do the DiffTRe authors see as the main limitation of that kind of training, and what do they train on instead?
Reference answer. SchNet is trained bottom-up on DFT reference calculations (PBE+vdW-TS for C20). DiffTRe's authors argue that bottom-up training is limited by the availability of precise and broad enough quantum data sets, and that its accuracy is capped by the reference method. DiffTRe trains top-down instead, so that simulations reproduce experimental observables.
Evidence (one passage per part). Schütt et al. (SchNet) [11]: "By training SchNet on DFT data at the PBE+vdWTS level, we reduce the computation time per single point by three orders of magnitude"; Thaler & Zavadlav (DiffTRe) [13]: "Thus, the main obstacle in bottom-up learning of NN potentials is currently limited availability of highly precise and sufficiently broad data sets."

7 **q043** COMPARISON · `alphafold`, `nequip`

Question. AlphaFold's invariant point attention and NequIP both have to deal with rotations of the whole system. Which quantities does each keep invariant, and which equivariant?
Reference answer. In AlphaFold's IPA, query, key and value points are produced in each residue's local frame, so the attention output is invariant to global rotations and translations; the frame updates are then applied in each local frame, which makes the block as a whole equivariant on the residue gas. NequIP keeps its internal features equivariant (geometric tensors under rotation and reflection) throughout message passing, and only its output, the potential energy, is invariant.
Evidence (one passage per part). Jumper et al. (AlphaFold) [**?**]: "The IPA augments each of the usual attention queries, keys and values with 3D points that are produced in the local frame of each residue such that the final value is invariant to global rotations and translations"; Batzner et al. (NequIP) [1]: "ven though the output of NequIP is the predicted potential energy Epot, which is invariant under translations, reflection, and rotations, the network contains internal features that are geometric tensors which are equivariant to rotation and reflection. This constitutes the core difference between NequIP and existing . . . "

7 **q044** COMPARISON · `alphafold`, `difftre`

Question. AlphaFold and DiffTRe both face a shortage of labelled training data. What does each use to get around it?
Reference answer. AlphaFold adds unlabelled sequences through self-distillation: its own high-confidence predicted structures for about 350,000 Uniclust30 sequences become extra training data. DiffTRe sees the limit of bottom-up training as the lack of precise and broad enough quantum data sets, and instead trains top-down on experimental observables through reweighted simulations.
Evidence (one passage per part). Jumper et al. (AlphaFold) [**?**]: "we use a trained network to predict the structure of around 350,000 diverse sequences from Uniclust3036 and make a new dataset of predicted structures filtered to a high-confidence subset"; Thaler & Zavadlav (DiffTRe) [13]: "Thus, the main obstacle in bottom-up learning of NN potentials is currently limited availability of highly precise and sufficiently broad data sets."

7 **q055** COMPARISON · `numeric`, `esmfold`, `alphafold`

Question. For a 384-residue protein on a V100, how does ESMFold's prediction time compare with AlphaFold's own timings?
Reference answer. ESMFold takes 14.2 seconds, which the ESM paper calls 6× faster than a single AlphaFold2 model. The AlphaFold paper reports 9.2 minutes with ensembling and 1.1 minutes without, and ESMFold also skips the MSA search, which can take over 10 minutes on its own.
Evidence (one passage per part). Lin et al. (ESMFold) [**?**]: "On a single NVIDIA V100 GPU, ESMFold makes a prediction on a protein with 384 residues in 14.2 seconds, 6x faster than a single AlphaFold2 model."; Jumper et al. (AlphaFold) [**?**]: "Without ensembling, the network is 8× faster and the representative timings for a single model are 0.6 min with 256 residues, 1.1 min with 384 residues and 2.1 h with 2,500 residues."

7 q057 COMPARISON · `esmfold`, `alphafold`

Question. Both AlphaFold and ESMFold add predicted structures to their training data. How does each do it?

Reference answer. AlphaFold self-distils: its own model predicts ˜350,000 Uniclust30 sequences, a high-confidence subset is kept, and the final model samples 75% from these predictions and 25% from the PDB. ESMFold distils from AlphaFold2 instead: ˜13.5M AlphaFold2 predictions filtered to mean pLDDT > 70 (removing only 1.5%), no loss on residues with pLDDT < 70, also sampled 75% of the time against 25% real structures.

Evidence (one passage per part). Jumper et al. (AlphaFold) [**?**]: "we use a trained network to predict the structure of around 350,000 diverse sequences from Uniclust3036 and make a new dataset of predicted structures filtered to a high-confidence subset"; Lin et al. (ESMFold) [**?**]: "Predicted structures are sampled from 75% of the time, and real structures 25% of the time during training."

Multi-hop questions (4)

Two passages, where finding the second depends on the first.

7 q017 MULTIHOP · `schnet`, `nequip`

Question. Which kernel method did the SchNet paper compare against on MD17, and how does NequIP do against kernel methods of that kind?

Reference answer. SchNet compared against GDML (gradient-domain machine learning), which was more data-efficient on the small training set for most molecules. The NequIP paper reports outperforming kernel-based approaches, including sGDML (the symmetrized successor of GDML), as well as GAP and FCHL19.

Evidence (one passage per part). Schütt et al. (SchNet) [11]: "We also list the performances of gradient domain machine learning (GDML)21 and DTNN28 for reference."; Batzner et al. (NequIP) [1]: "We find that NequIP significantly outperforms invariant GNN-IPs (such as SchNet [9] and DimeNet [11]), shallow neural networks (such as ANI [40]), and kernel-based approaches"

7 q030 MULTIHOP · `difftre`, `behler-parrinello`

Question. DiffTRe's diamond model starts from an empirical potential originally built for another element. Which potential was it, and which neural network potential paper used that same element as its test system?

Reference answer. The Stillinger-Weber potential parametrized for silicon, rescaled for carbon. Silicon is also the test system of the original Behler-Parrinello neural network potential, which was compared against empirical potentials and DFT for bulk silicon.

Evidence (one passage per part). Thaler & Zavadlav (DiffTRe) [13]: "To demonstrate learning, we select the original Stillinger-Weber potential [46] parametrized for silicon as Uprior."; Behler & Parrinello [2]: "The high accuracy of the method is demonstrated for bulk silicon and compared with empirical potentials and DFT"

7 q031 MULTIHOP · `difftre`, `nequip`

Question. Which graph neural network does DiffTRe use as its learnable potential, and how does NequIP compare with that family of models on MD-17?

Reference answer. DimeNet++ (an improved DimeNet), used for both diamond and coarse-grained water. NequIP reports significantly outperforming invariant graph network potentials, naming SchNet and DimeNet, on MD-17.

Evidence (one passage per part). Thaler & Zavadlav (DiffTRe) [13]: "Second, we train the state-of-the-art graph neural network potential DimeNet++ [11, 12] for an atomistic model of diamond from its experimental stiffness tensor."; Batzner et al. (NequIP) [1]: "We find that NequIP significantly outperforms invariant GNN-IPs (such as SchNet [9] and DimeNet [11])"

7 q056 MULTIHOP · `esmfold`, `alphafold`

Question. ESMFold's folding block is adapted from an AlphaFold block. Which one, and what had to change because there is no MSA?

Reference answer. The Evoformer. AlphaFold's Evoformer keeps a 2D MSA representation (axial attention over its rows and columns) and updates the pair representation from it through an outer product summed over the sequences. ESMFold's language-model features are one-dimensional, so axial attention becomes standard self-attention over the sequence (still biased by the pair representation), sequence and pair communicate

through an outer product and an outer difference, and templates are replaced by the language model's attention maps.
Evidence (one passage per part). Lin et al. (ESMFold) [**?**]: "Since MSAs are two dimensional, the Evoformer employs axial attention (69) over the columns and rows of the MSA. The language model features are one dimensional, so we can replace the axial attention with a standard attention over this feature space."; Jumper et al. (AlphaFold) [**?**]: "The MSA representation updates the pair representation through an element-wise outer product that is summed over the MSA sequence dimension."

Unanswerable questions (5)

The answer is not in the corpus; the right response is to decline.

7 **q021** UNANSWERABLE · `out-of-scope`
Question. What experimental success rate does RFdiffusion reach when designing protein binders?
Reference answer. Not answerable from this corpus, which covers interatomic potentials and force fields, not protein design.

7 **q022** UNANSWERABLE · `schnet`
Question. What dropout rate was used when training SchNet?
Reference answer. Not in the corpus. The SchNet paper describes its optimizer, batch size and learning-rate decay, but no dropout.

7 **q033** UNANSWERABLE · `difftre`
Question. How many GPU hours did it take in total to train the DiffTRe diamond potential?
Reference answer. Not in the corpus. The main text gives only per-update wall-clock times for the toy example and refers to the supplementary information for the diamond training curve; no total training cost is stated.

7 **q045** UNANSWERABLE · `alphafold`
Question. What learning rate and optimizer were used to train AlphaFold?
Reference answer. Not answerable from this corpus: the paper only says fine-tuning used a reduced learning rate and refers to Supplementary Methods 1.11 for the exact configuration, which is not part of the PDF.

7 **q058** UNANSWERABLE · `esmfold`
Question. What TM-score does ESMFold reach on the CASP15 targets?
Reference answer. Not answerable from this corpus; the ESMFold paper evaluates on CASP14 and CAMEO, not CASP15.

C AI Usage

D Old results

D.1 Retrieval

D.2 End-to-end answers

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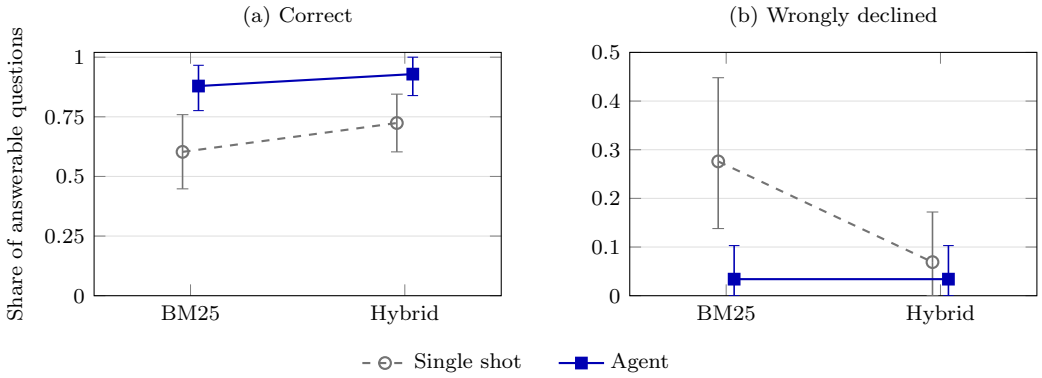


Figure 3: Answer quality for single shot and the agent with BM25 and hybrid retrieval (Claude Sonnet 5.5, 29 answerable dev questions, judged by Claude Sonnet 5.5). Points are means, bars 95% bootstrap intervals of each run. (a) The agent improves correctness with either retriever; hybrid retrieval helps single shot more than the agent. (b) With BM25, much of single shot's deficit is declining questions the corpus does answer, when its one search misses the evidence; hybrid retrieval removes most of these, and the agent's remaining advantage comes from answering better rather than more often. Paired comparisons are in Table 6.

	Correct			Wrongly declined		
Comparison (B vs. A)	Δ [95% CI]	B/W	p	Δ [95% CI]	B/W	p
Agent vs. single shot, BM25	+0.28 [+0.12, +0.43]	10/1	0.012	−0.24 [−0.41, −0.10]	7/0	0.016
Agent vs. single shot, hybrid	+0.20 [+0.09, +0.32]	9/0	0.004	−0.03 [−0.10, 0.00]	1/0	1.000
Hybrid vs. BM25, single shot	+0.12 [−0.02, +0.26]	8/2	0.109	−0.21 [−0.34, −0.07]	6/0	0.031
Hybrid vs. BM25, agent	+0.04 [−0.04, +0.11]	3/1	0.625	0.00 [0.00, 0.00]	0/0	1.000
Agent + hybrid vs. single shot + BM25	+0.32 [+0.16, +0.48]	11/0	0.001	−0.24 [−0.41, −0.10]	7/0	0.016

Table 6: Paired comparisons on the same questions (n = 28–29 answerable dev questions). Δ is the mean per-question difference B − A with a paired bootstrap 95% interval; B/W counts the questions on which B did better / worse than A (for wrongly declined, "better" means it declined less); p is a two-sided exact sign test on those counts. Correctness is graded by Claude Sonnet 5.5 against the reference answers.

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